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Compensation and performance: A review and recommendations for the future

Ingrid Smithey Fulmer[1] | Barry Gerhart[2] | Ji Hyun Kim[3]

[1]School of Labor and Employment Relations, University of Illinois at Urbana-Champaign, Champaign, Illinois, USA

[2]Department of Management and Human Resources, Wisconsin School of Business, University of Wisconsin-Madison, Madison, Wisconsin, USA

[3]Department of Management and Organisation, NUS Business School, National University of Singapore, Singapore, Singapore

Correspondence
Ingrid Fulmer, School of Labor and Employment Relations, University of Illinois at Urbana-Champaign, 504 E. Armory Avenue, Champaign, IL 61820, USA.
Email: if85@illinois.edu

Abstract

In this paper, we undertake two primary tasks. First, we describe trends in pay → performance research since 1990. Examining 30+ years allows us to capture the evolution of thinking and methodology over time. For this purpose, we focus on scholarly work appearing in *Personnel Psychology (PP)*, as well as in the *Journal of Applied Psychology* (*JAP*) and *Academy of Management Journal* (*AMJ*). As our second task, we go beyond analysis of broad trends, to take a deeper and more substantive dive into the specific topic of pay for performance (PFP) and related areas (pay dispersion, pay communication/transparency), while at the same time drawing on the broader literature on PFP that goes beyond the three journals listed above. We also discuss pay equity, endogeneity, and international issues. Finally, we suggest practical implications, as well as future research directions.

KEYWORDS
compensation, pay-for-performance, performance

1 | INTRODUCTION

The payment of compensation in exchange for the contributions provided by employees is uniquely important in defining the employment relationship. Indeed, the very definition of the word "employ" ("to provide with a job that pays wages or salary," Merriam-webster.com) implies that compensation is both necessary and sufficient to define the relationship. Other practices or social exchanges in organizations may be nice to have, wise to have, or even legally required, but are not relationship-defining in the way that compensation is (Rousseau & Ho, 2000). The inducements (e.g., employers offering pay) and contributions (e.g., employees providing performance) defining this exchange form a contract, implicit and/or explicit, with different varieties of contracts/employment relationships in existence (Adams, 1963; Barnard, 1968; Fulmer et al., 2003; March & Simon, 1958; Rousseau, 1989; Tsui et al., 1997; Williamson, 1975).

Compensation includes "any direct or indirect payments to employees, such as wages, bonuses, stock, and benefits" (Gerhart & Milkovich, 1992, p. 484). From an organization's point of view, successful achievement of performance goals requires offering compensation at a level and in a form that attracts, retains, and motivates a workforce having the desired attributes. At the same time, it must carefully manage compensation costs, which are often the single largest operating cost. For example, U.S. employer costs for employee compensation in the civilian sector in early 2022 (Bureau of Labor Statistics, 2022) averaged $40.90/hour, with $28.16 in the form of wages, salaries, and other direct pay, and $12.74 ($2.94 legally required) in the form of benefits. Although most organizations do not report labor cost data in their public filings, there are exceptions in the airlines, healthcare, and financial services industries. A sampling indicates that compensation accounts for 30 to 58% of operating expenses and 28 to 46% of revenues (Noe et al., 2023; Table 11.1).

For most employees, compensation received via the employment relationship is their major source of income and instrumental for achieving so many kinds of needs and/or goals (e.g., security, status, esteem, achievement; Lawler, 1971), or stated differently, in maintaining the overall health and wellbeing of one's self and one's family outside of work (Leana & Meuris, 2015). Pay is also universally valued, with more pay generally preferred to less (Rottenberg, 1956), unlike other job attributes (e.g., responsibility, degree of supervision), whose attractiveness depends on the employee (Dawis et al., 1968; Lawler, 1971; Schneider, 1987; Weller et al., 2019). Although some have suggested that once pay rises above a certain level, well-being plateaus (e.g., at around $75,000 in 2010, based on one interpretation of Kahneman & Deaton, 2010), more recent research has found, based on 1,725,994 experience-sampling reports from 33,391 employed US adults, that well-being "increased linearly with log(income), with an equally steep slope for higher earners as for lower earners" and that there "was no evidence for an experienced well-being plateau above $75,000/y, contrary to some influential past research." (Killingsworth (2021, p. 1).

Employees evaluate and react to their compensation within a social context where perceptions of fairness/equity are central. From a societal viewpoint, workers' compensation has important implications for national productivity and thus standards of living. Societal considerations can also be seen in the extensive regulatory framework for compensation in many developed countries (e.g., equal employment opportunity, minimum wage and overtime provisions), the continued focus of labor unions on compensation issues, and the significant attention to income inequality in both developed and developing countries around the world.

An extensive literature also documents the importance of pay in terms of how it affects what "people do" (Rynes et al., 2004). This includes findings that higher pay levels substantially reduce turnover (Gerhart, 2023, Exhibit 17.15), as well as that there is a significant increase in pay that follows (voluntary) turnover (Gerhart, 2023, Exhibit 17.16), especially among those with high human capital (e.g., 35% in Bidwell, 2015). Of course, the evidence (e.g., meta-analytic evidence provided by Jenkins et al., 1998; Kim et al., 2022) including work that we review later, also shows substantial positive effects of PFP on worker performance. Increasingly, scholars are also paying attention to how pay systems affect performance, turnover and other outcomes in different ways for different people, depending on characteristics such as individual differences, demographics, and relative workgroup standing (Fulmer & Shaw, 2018).

Accordingly, research on compensation is scientifically and practically important and has been published in the pages of *Personnel Psychology* (*PP*) from its inception. In the introduction to the very first issue of *PP*, editors Erwin Taylor and Charles Mosier listed the types of research questions of relevance to the journal, including: "What are the techniques for establishing wage policies and wage rates that will result in an equitable wage structure, psychologically acceptable to those who work under it?" (1948, p. 3), as well as the related topic of workers' motivations, morale, and attitudes and the effectiveness of methods for improving them. Later in that year, the first two compensation-related articles were published in *PP* (Cohen, 1948; Jurgensen, 1948). Fast-forwarding to today, this seventy-fifth anniversary of *PP* marks an excellent time to take stock and consider the state of contemporary employee compensation research, in this journal as well as more broadly. With over fifty-seven (!) years of combined interest and experience studying and writing about compensation, we are excited about this opportunity to reflect on where we've been and to offer suggestions for future directions.

In this paper, we undertake two primary tasks. First, we describe trends in research on pay → performance since 1990. Examining 30+ years allows us to capture the evolution of thinking and methodology over time. One reason we chose to begin with the year 1990 is that two comprehensive reviews, one on compensation broadly (Gerhart & Milkovich, 1992) and one on pay for performance (PFP) specifically (Milkovich & Wigdor, 1991) cover the literature up to approximately 1990. A second reason is that 1990 is roughly when the compensation literature began to move from a mostly individual (and sometimes team) level of analysis to business unit and organization levels, including strategic perspectives focusing on sustained organization differences in compensation practices and their consequences for organization/unit performance (Balkin & Gomez-Mejia, 1990; Gerhart & Milkovich, 1990; Gomez-Mejia & Balkin, 1992), with this supra-individual perspective eventually broadening to other individual human resource practices (e.g., staffing, Terpstra & Rozell, 1993) and then to systems of human resource practices (e.g., Arthur, 1994; Huselid, 1995; MacDuffie, 1995; see Becker & Gerhart, 1996 for an overview of this shift).

For this purpose, we focus on scholarly work appearing in *PP*, as well as in the *Journal of Applied Psychology* (*JAP*) and *Academy of Management Journal* (*AMJ*). Because covering every important topic in detail is not feasible, we narrowed our focus to research where performance is the dependent (endogenous) variable, since "the ultimate dependent variable when making strategic pay choices is . . .performance" (Gomez-Mejia et al., 2014, p. 58). On the independent (exogenous) variable side, we also narrow our focus to direct pay and here we will see that PFP is the most frequently studied aspect of direct pay. We will also see that performance as both a dependent variable and as part of a PFP independent variable can be measured at one or more distinct levels (e.g., individual, unit, organization) and with results- and/or behavior-based performance measures. The performance measures on the two sides of the equation can be the same (e.g., a PFP plan where individual incentive payouts depend on individual productivity results, with the goal of increasing individual productivity results) or different (e.g., a PFP plan where individual base pay increases dependent on individual performance behavior ratings, with the goal of increasing organization profits). Although other compensation elements (e.g., employee benefits) are of course important, fortunately, these have been the subject of other recent reviews (e.g., Fulmer & Li, 2022) and/or are covered in detail in book-length treatments (Gerhart, 2023).

As our second task, we go beyond analysis of broad trends, to take a deeper and more substantive dive into the specific topic of PFP and related areas (pay dispersion, pay communication/transparency), while at the same time drawing on the broader literature on PFP. One major reason for our focus on PFP is that, as we will see (Tables 1 and 2), PFP has been the focus of the vast majority of compensation research in *PP*, *JAP*, and *AMJ* that uses performance as the outcome variable. Another reason is the longstanding argument that PFP is the most strategic dimension of compensation strategy. Gerhart and Milkovich (1990) noted that labor market and product market competition set a floor and ceiling, respectively, on pay level (i.e., average pay), often limiting firms' ability to differentiate based on how much they pay (which they might like to do because of the high importance of pay level and how visible/quantifiable it is), but not in how they pay—hence the popularity of PFP (See Gerhart & Rynes, 2003, p. 21 and Haire et al., 1967, p. 9–10 for similar arguments. See Rahmandad & Ton, 2020 for an example of different within-industry pay level strategies.).

Understanding the current state of and trends in empirical research on employee compensation and performance (our first task), and key findings on PFP specifically (our second task) is of critical theoretical and practical, bottom-line importance. If research finds certain pay systems are less effective than expected, this might be because of design flaws, poor execution (e.g., poor communication), or lack of alignment with the workplace context. Identifying the reason is important in guiding steps to a remedy. Methodological soundness also matters because research lacking validity should not inform compensation decisions. We conclude by offering several specific practical recommendations for managers that emerge from our review, and identifying avenues that we believe will be fruitful for future scholars to traverse.

2 | TASK 1: TRENDS IN COMPENSATION RESEARCH

2.1 | Selecting articles

We first searched *Personnel Psychology* (*PP*), *Journal of Applied Psychology* (*JAP*), and *Academy of Management Journal* (*AMJ*) using Wiley Online Library to identify articles containing the keywords *compensation*, *pay*, *reward*, *bonus*, *incentive*, or *salary* in the title or abstract, published since 1990. Second, given space and time constraints, we narrowed our substantive focus by including only articles where: (a) performance was the endogenous variable; (b) compensation (in terms of direct pay) was the exogenous variable (or moderator); (c) the focus was on employees other than the top 5 executives or boards; and (d) the relationship of compensation with performance could be identified separately. We excluded studies using turnover as the endogenous variable unless they specifically examined turnover of high performers.

2.2 | Examining trends

Table 1 shows characteristics of each study and Table 2 provides a summary. Since 1990, 39 articles examined compensation → performance, with 13 in the 1990s, 15 in the 2000s, declining to 11 since 2010 (despite being a 13- rather than 10-year interval). These numbers are consistent with observations of previous researchers who have noted the scarcity of compensation research, both in absolute terms and relative to the volume of research on other HR practices such as employee selection and performance appraisal (e.g., Gupta & Shaw, 2014). Examination of Table 1 also shows that although a range of compensation constructs have been studied, PFP and closely related topics (e.g., PFP perceptions, PFP exceptions/i-deals) has been by far the most studied (in 26 of 39 studies, with the next most studied topics being covered in 3 of 39 studies).

With respect to theories, Table 2 shows expectancy theory, based in psychology, and agency theory, based in economics, have been used most often (seven and six studies each, respectively), followed by equity, goal-setting and tournament/sorting theories (five studies each). Also of note are three other theories used twice each, plus another 15 theories used once each (included in the "others" column); a few studies have used multiple theories. Overall, there is much theoretical diversity, with a few theories used more often than others, but far from dominant. In the most recent (2010-2022) period, however, only two of the frequently used theories (equity and expectancy) were used more than once. Agency and goal-setting, more commonly occurring theories in the past, have disappeared from recent pay → performance research (and tournament theory/sorting nearly so). For interested readers, reviews of the theories above are available from a number of sources (e.g., Bartol & Locke, 2000; Gerhart, 2023 [see Exhibit 9.4 for a brief summary of theories' essential features and implications for PFP]; Gerhart & Rynes, 2003). We also note that some studies in Table 1 (e.g., Petty et al., 1992) have placed little emphasis on delineating and testing theory, taking a primarily applied focus.

PFP programs are typically classified on two dimensions (Gerhart & Rynes, 2003; Gerhart et al., 2009): (1) use of results-oriented (e.g., physical production output, sales, profits, total shareholder return) or behavior-oriented (e.g., supervisor or customer ratings of specific behaviors) performance measures; and (2) use of performance measured at the individual or aggregate (e.g., group, unit, organization) level.[i] (See Nyberg et al., 2018 for a review of aggregate pay plans.) A third dimension, not included in this section due to it being infeasible to code for most studies (Kim et al., 2022), but which we return to later, is incentive intensity (magnitude of the pay-performance relationship, Gerhart & Rynes, 2003). Because "incentive" is often also used to refer specifically to individual incentive plans, we will use the term PFP intensity. We will say more later regarding the effectiveness implications of these PFP decisions.

Table 2 shows that both results- (27 studies) and behavior-based (11 studies) measures of performance are commonly used, but results-based measures, which often lend themselves more easily to research (e.g., in the form of

TABLE 1 Theories, compensation constructs, and selected methods to study pay → performance, by time period, in PP, JAP, and AMJ

	Study	Compensation construct	Performance dependent variable and type	Level of analysis	Theory	Endogeneity: Design approach	Endogeneity: Statistical approach
1990-1999 (*N* = 13)							
	Gerhart & Milkovich (1990, AMJ) [B]	Pay level & PFP long term incentives, bonus [IV]	ROA [R]	Organization	Expectancy, agency	Longitudinal (archival)	Fixed effects model
	Petty et al. (1992, JAP) [W]	PFP general [IV]	Sales, expense [R]	Unit		Quasi experiment	
	Wright et al. (1993, JAP)	PFP bonus [IV]	Quantity produced [R]	Individual	Goal setting	Lab experiment	
	Banker et al. (1996, AMJ) [W]	PFP bonus [IV]	Sales, profit [R]	Unit	Organizational control, agency	Quasi experiment	
	Erez and Somech (1996, AMJ) [I]	PFP bonus [IV]	Quantity produced [R]	Individual	Self-regulation, social loafing, goal setting	Lab experiment	
	Klaas and McClendon (1996, PP) [B]	Pay level [IV]	Financial impacts [R]	Organization	Utility	Simulation	
	Roth & O'Donnell (1996, AMJ) [I, II] [B]	Pay level (perception) [IV]	Perceived effectiveness [B]	Unit	Agency		
	Stewart (1996, JAP) [W]	PFP commission [MV]	Number of customer memberships [R]	Individual	Behavioral activation theory	Quasi experiment	

(Continues)

TABLE 1 (Continued)

	Study	Compensation construct	Performance dependent variable and type	Level of analysis	Theory	Endogeneity: Design approach	Endogeneity: Statistical approach
	Trevor et al. (1997, JAP) [W]	PFP salary raise & promotion [MV]	Turnover quality using performance rating [B]	Individual	March & Simon turnover model	Longitudinal (archival)	
	Bloom & Milkovich (1998, AMJ) [B]	PFP bonus [IV]	Shareholder return [R]	Multi-level	Agency	Longitudinal (archival)	
	Murray and Gerhart (1998, AMJ) [W]	Skill-based pay [IV]	Labor cost, defective rates, labor productivity [R]	Unit	Expectancy, job design	Longitudinal (archival)	
	Bloom (1999, AMJ) [B]	Pay dispersion [IV]	Baseball player and team performance [R]	Multi-level	Neoclassical economics	Longitudinal (archival)	
	Wood et al. (1999, JAP)	PFP bonus [IV]	Production hours [R]	Individual	Goal setting	Lab experiment	
2000-2009 *(N = 15)*							
	Knight et al. (2001, AMJ)	PFP bonus [IV]	Computer game score [R]	Unit	Goal setting, social cognitive, agency, prospect	Lab experiment	
	Stajkovic and Luthans (2001, AMJ) [W]	PFP general [IV]	Production hours [R]	Individual	Social cognitive	Quasi experiment	
	Beersma et al. (2003, AMJ)	PFP incentive [IV]	Speed & accuracy in computer simulation [R]	Multi-level	Goal interdependence theory	Lab experiment	

(Continues)

TABLE 1 (Continued)

Study	Compensation construct	Performance dependent variable and type	Level of analysis	Theory	Endogeneity: Design approach	Endogeneity: Statistical approach
Brown et al. (2003, AMJ) [B]	Pay level & pay dispersion [IV]	ROA, patient length of stay, survival rate [R]	Organization	Efficiency wage, equity	Longitudinal (archival)	
Sturman et al. (2003, PP) [B]	PFP incentive [IV]	Turnover quality using performance rating [B]	Multi-level	Utility	(Simulation)	
Schweitzer et al. (2004, AMJ)	PFP bonus [MV]	Unethical behavior [B]	Individual	Goal setting	Lab experiment	
Currall et al. (2005, PP) [B]	Pay satisfaction (perception) on pay structure, benefit, raise, and level [IV]	Student educational outcomes [R]	Organization	Equity		
Russo and Harrison (2005, AMJ) [W]	PFP general [IV]	Toxic emission [R]	Unit		Quasi experiment	
Salamin & Hom (2005, JAP) [I] [W]	PFP merit, bonus, promotion [MV]	Turnover quality using performance rating [B]	Multi-level	Tournament, sorting	Longitudinal (archival)	
Combs et al. (2006, PP) [B]	Pay in HPWS; incentive, pay level, internal promotion [IV]	Productivity, retention, returns, growth [R]	Organization	HPWS theory		
Morgeson et al. (2006, PP) [W]	PFP general (perception) [MV]	Effort expended [B]	Multi-level	Job design	Quasi experiment	
Peterson and Luthans (2006, JAP) [W]	PFP bonus [IV]	Gross profit, production time, turnover rate [R]	Unit	Self-regulation	Quasi experiment	

(Continues)

TABLE 1 (Continued)

	Study	Compensation construct	Performance dependent variable and type	Level of analysis	Theory	Endogeneity: Design approach	Endogeneity: Statistical approach
	Cadsby et al. (2007, AMJ) [I]	PFP piece rate [IV]	Quantity of anagram completed [R]	Individual	Agency, expectancy, sorting	Lab experiment	
	Shaw and Gupta (2007, PP) [B]	Pay dispersion [IV MV]	Turnover quality using performance rating [B]	Multi-level	Tournament, sorting	Longitudinal (archival)	
	Kepes et al. (2009, PP) [B]	Performance- vs. politically-based pay (perception) & pay range [IV MV]	Productivity; accident frequency ratio, out-of-service percentage [R]	Organization	Tournament, expectancy	Longitudinal (archival)	
2010-2022 *(N = 11)*							
	Bamberger and Belogolovsky (2010, PP) [I]	Pay communication (secrecy vs. transparency) [IV]	Computer simulation score [R]	Individual	Expectancy, equity	Lab experiment	
	Chuang and Liao (2010, PP) [I B]	Pay in HPWS; pay level, PFP general, benefits, fairness of pay (perception) [IV]	Service performance, helping behavior, market performance [R B]	Multi-level	HPWS theory, organizational climate theory	Longitudinal (archival)	
	Trevor et al. (2012, AMJ) [B]	Pay dispersion [IV]	National Hockey League performance [R]	Organization	Equity, sorting	Longitudinal (archival)	Instrumental variable analysis
	Kownatzki et al. (2013, AMJ) [I II B]	PFP bonus, profit sharing, benefits, career prospects (perception) [IV]	Decision speed [B]	Multi-level	Organizational control		

(Continues)

TABLE 1 (Continued)

Study	Compensation construct	Performance dependent variable and type	Level of analysis	Theory	Endogeneity: Design approach	Endogeneity: Statistical approach
Belogolovsky & Bamberger (2014, AMJ) [I]	Pay communication (secrecy vs. transparency) [IV]	Computer simulation score [R]	Individual	Signaling	Lab experiment	Fixed effects model
Han et al. (2015, JAP) [I B]	PFP general (perception) [IV MV]	Performance rating of individual performance [B]	Individual	Expectancy		
Maltarich et al. (2017, AMJ) [W]	PFP commission [IV]	Revenue [R]	Individual	Economics, expectancy, psychological contract theory	Longitudinal (archival)	
Abdulsalam et al. (2021, JAP) [W]	PFP i-deals (perception) [IV]	Sales [R]	Individual	Social comparison, equity	Longitudinal (archival)	
Han et al. (2021, PP) [I W]	Reward interdependence (perception) [MV]	Sales [R]	Unit	Social hierarchy	Longitudinal (archival)	
SimanTov-Nachlieli & Bamberger (2021, JAP) [I II]	Pay communication (perception) [IV]	Work behavior [B]	Individual	Uncertainty management	Lab experiment	
Kong et al. (in press, AMJ) [I W]	PFP general (perception) [IV]	Performance rating of individual performance [B]	Individual	Transactional theory of stress, role engagement	Longitudinal (archival)	

[I]Includes at least one non-U.S. sample. [II] Includes samples from two or more countries. [B] Between-organization design. [W] Within-organization design. [IV] Compensation as independent variable. [MV] Compensation as moderator. [R] Results-based performance measure. [B] Behavior-based performance measure.

TABLE 2 Use of theories and selected methods to study pay → performance, by time period, in PP, JAP, and AMJ

Summary count	Theory[1]									Performance measure		
	Agency	Equity	Expectancy	Goal setting	Job design	Social cognitive	Tournament/ Sorting	Utility	Others[2]	Results	Behavior	Both
1990-1999	4	0	2	3	1	0	0	1	4	11	2	0
2000-2009	2	2	2	2	1	2	4	1	3	10	5	0
2010-2022	0	3	3	0	0	0	1	0	8	6	4	1
Total	*6*	*5*	*7*	*5*	*2*	*2*	*5*	*2*	*15*	*27*	*11*	*1*

	Level of analysis				Endogeneity: Design approach			Endogeneity: Statistical approach			
	Individual	Unit	Organization	Multi-level	Longitudinal (archival)	Lab experiment	Quasi experiment	Fixed effects model	Instrumental variable	Selection bias	*Grand Total*
1990-1999	5	4	2	2	5	3	3	1			*13*
2000-2009	3	3	4	5	4	4	4			1	*15*
2010-2022	7	1	1	2	6	3	0	1	1		*11*
Total	*15*	*8*	*7*	*9*	*15*	*10*	*7*	*2*	*1*	*1*	*39*

[1]A few studies used multiple theories, making the sum of the theories used larger than the number of studies. For instance, expectancy and agency theories are separately counted for Gerhart and Milkovich (1990).

[2]Includes efficiency wage, self-regulatory, social loafing, March & Simon turnover model, organizational control, neoclassical economics, goal interdependence theory, prospect, HPWS, organizational climate, signaling, psychological contract, social comparison, transactional theory of stress, role engagement, uncertainty management, and social hierarchy theories.

individual incentives, which have frequently been reviewed in meta-analyses), are used about twice as often. In the most recent time period, however, results and behavior measures were used almost equally.

The individual (15 studies) and aggregate (unit, organization; 15 studies combined) levels of analysis are used equally, but in the most recent time period (2010-2022), there has been a shift to use of mostly individual level data. As another observation, we have long known that aggregating individual level relationships such as PFP → performance does not necessarily reproduce individual-level statistical relationships (e.g., Ostroff, 1993; Robinson, 1950). However, individual-level relationships play a key role in the aggregate level PFP → performance relationship and indeed this has become a focus of attention in the strategy literature under the term "microfoundations" (Felin et al., 2015). For PFP research to provide insight into these issues, use of multi-level data is necessary. However, we have seen only two such studies in the most recent time period.

Another design feature we show in Table 1 (see B and W superscripts in column 1) that we think aspiring compensation researchers should be aware of is whether a study is between-organization or within-organization. The former presents more of a challenge if collecting primary data (which is not always necessary—e.g., consulting firm data have been used in such research), whereas the latter presents more of a (potential) challenge in assuring that variation in compensation practices within a single organization is sufficient.

The remaining columns in Table 1 summarize how research has dealt with endogeneity challenges. Much attention has been devoted to endogeneity using a design approach (e.g., longitudinal data, experiments, quasi-experiments). Although use of laboratory experiments seems to have remained fairly consistent over the decades, it appears that quasi-experiments (e.g., Petty et al., 1992), seen in the 1990s and 2000s, have not been used since 2010. Another noteworthy takeaway from Table 2 is that statistical approaches to address endogeneity are almost never used, regardless of the decade. Indeed, Table 2 indicates only 4 such instances in 39 articles over 30 years. When endogeneity has already been addressed to some degree through design, especially in the case of an experiment, this is more understandable. However, almost all experiments are conducted in the laboratory. Field research in compensation, at best, makes use of quasi-experimental designs (which as noted have not been done of late), which are unlikely on their own to fully address endogeneity concerns. Clearly, there is room for future research to better address endogeneity. We will return to this topic later.

Finally, returning to column 1 in Table 1 (see I and II superscripts in column 1), we see that 27 studies used only U.S. samples. Three studies used multiple countries, one of which was comparative (using cultural distance between each country and the headquarters country; Roth & O'Donnell, 1996).

3 | TASK 2: FOCUS ON PFP→ PERFORMANCE

To this point, we have focused on trends in compensation research examining pay → performance in three journals over a 30+ year timeframe. We now shift our focus in two ways. First, we narrow our substantive focus to how PFP influences performance. As we saw in Table 1, this has been by far the most studied topic. Further, emphasis on PFP → performance is consistent with our earlier discussion of the literature's emphasis on PFP as an especially strategic compensation dimension. Second, we expand our focus to include literature beyond *PP*, *JAP*, and *AMJ*. However, many of the individual studies we highlight continue to come from these three journals. In examining PFP → performance, we include closely related areas such as performance measurement/distribution, pay transparency, and pay dispersion. Based on our examination of trends above, we address the need for more attention to endogeneity and international issues. Finally, we address diversity/inclusion issues, which in compensation typically means examining pay equity. We begin with a brief review of beliefs about the role performance should have in determining pay and by describing two broad conceptual mechanisms (incentive and sorting effects) that capture many of the more specific and varied mechanisms in the wide array of theories (Tables 1 and 2) used in PFP → performance research.

3.1 | Employee and employer views on the role of performance in pay (and its use)

What role should performance have in determining pay? Classical research, nearly all in *PP*, has shown that employees and managers largely agree that pay should be based on merit/performance. For example, 180 managers from 72 different (U.S.) companies rated nine possible factors in terms of the importance they should receive in determining the size of employees' salary increases; job performance was deemed most important, followed by the nature of job and amount of effort expended (Dyer et al., 1976). Research by Fossum and Fitch using U.S. data (1985) also found performance to be most important and rated considerably higher than length of service. Similar findings are reported in other studies using not only U.S. data, but also data from China and Russia (Giacobbe-Miller et al., 1998; Sherer et al., 1987; Zhou & Martocchio, 2001). In addition, Williams, McDaniel, and Nguyen's (2006) meta-analysis found employees' pay satisfaction positively related to their perception of the degree to which their organization uses PFP ($r = .31$).

Employers also believe in paying for performance, especially for more critical jobs. Virtually all private sector (and many other) organizations use PFP, usually in multiple forms (Gerhart, 2023), suggesting that employers believe pay is a key motivator. Even in unionized organizations where within-job pay increases may depend on seniority, advancement to higher job levels with higher pay typically also includes a performance element. In addition, we know that both pay level and proportion of PFP as a share of pay are greater in jobs at higher levels in the organization where contributions and performance are particularly critical to the broader business (Gerhart, 2023, Exhibit 10.2). These facts suggest that organizations believe that those making the most valuable contributions need/should receive higher pay in return.

3.2 | PFP → Performance fundamental mechanisms: Incentive and sorting effects

As we have seen, a wide range of theories have been used to study PFP. Here, to simplify, we describe PFP effects as operating via two broad conceptual mechanisms in these theories: incentive and/or sorting effects (Cadsby et al., 2007; Fulmer & Shaw, 2018; Gerhart & Fang, 2014; Gerhart & Rynes, 2003; Lazear, 1986). The incentive effect is the focus of all the most commonly used theories above. The incentive effect describes how PFP changes the performance of the current workforce, primarily through its impact on motivation of individual workers. The sorting effect, which is a more recent focus, operates by changing who the workers are, not by changing the behaviors of current workers. The classic study by Lazear (1986) examined workforce productivity (number of car windshields installed at Safelite) before and after implementation of an individual incentive. Productivity increased by 44%. However, productivity of workers present before and after implementation increased by 22%, leaving 44%–22% = 22% unexplained. Lazear found that the remaining 22% productivity increase came from lower productivity workers (whose relative pay decreased) becoming more likely to leave under the new PFP plan and to be replaced by higher productivity workers (who benefited most from the new PFP plan). He termed this a sorting effect.

Returning to Table 1 and 2, four studies examine and find support for sorting effects (Cadsby et al., 2007; Salamin & Hom, 1997; Shaw & Gupta, 2007; Trevor et al., 2012). Most notable perhaps is Cadsby et al. (2007), both because it introduced the use of a multi-stage (eight rounds) laboratory experiment to study sorting and because it included both conditions where subjects were assigned (regardless of their preferences) to PFP (vs. fixed rate) pay and conditions where subjects could choose PFP or fixed pay. Importantly, Cadsby et al. (2007) found that when given a choice, 35 of 115 switched from the pay condition initially (randomly) assigned. As a result, productivity went from being 10% higher in the PFP condition when conditions were (randomly) assigned to being 38% higher in the PFP condition after subjects could choose pay condition. This implies that sorting accounted for (38–10)/38 = 74% of the positive effect of PFP (See analysis in Gerhart, 2023, chapter 9.) That, in turn, means that 74% of the PFP effect would have been missed using a one-round between-subjects design with random assignment, because of the absence of pay condition choice and thus opportunity for "mobility" by subjects to their preferred pay condition.

Nevertheless, we have seen that the incentive effect continues to be by far the main focus, meaning that most research underestimates the effect of PFP that would be observed when employee mobility/sorting is included in the design. Bearing that in mind, we now turn to evidence on incentive effects (where sorting is not included). The "cleanest" form of evidence for the incentive effect comes from meta-analytic studies of individual incentive plans, where individual performance is measured as results (e.g., productivity) (e.g., Jenkins et al., 1998; Kim et al., 2022; Weibel et al., 2010). Interestingly, here, we have a case where Table 1 does not capture the large literature (in terms of primary studies, which in recent decades are mostly published elsewhere) on the effect of PFP using individual, results-based performance measures (i.e., individual incentives), whereas these meta-analyses do. Kim et al.'s (2022) meta-analysis, the most recent and complete, included 82 effect sizes and 7978 subjects/employees. Consistent with Jenkins et al. (1998), they found performance to be higher (by .54 SD) under incentive (vs. fixed/hourly) pay, which worked out to be a 24% performance difference.

Knowing the direction (positive or negative) and magnitude of the effect of any HR practice, including a compensation program, is of course important. However, to fully understand the (net) payoff, one must also incorporate the costs of the program. This is not common in the literature. One example, in the meta-analysis by Kim et al. (2022), found that the 24% performance improvement under incentives (vs. fixed/hourly pay) was still around 20% even after accounting for the fact that payouts were higher in incentive conditions than in fixed pay conditions. A model that can be used to incoporate a fuller set of costs and benefits is utility analysis (Brogden, 1949; Schmidt & Hunter, 1998). We are aware of only two studies in compensation that have used utility analysis, both appearing in *PP*. Klaas and McClendon (1996) found in a study of bank tellers that higher pay level increased employee performance, but not enough to cover the higher labor costs. Sturman et al. (2003) found in a study of exempt (i.e., professional and managerial) employees that although stronger PFP cost more, the resulting increase in performance (via sorting effects) and the overall net value created was greater. Note that both studies used a single employer and Klaas and McClendon (1996) studied bank tellers, a low wage position (and one with relatively low SDy in dollar terms). Thus, these studies demonstrate how to use utility analysis to study net compensation effects, although the specific results are not necessarily generalizabile.

One other issue that may prove to be critical in both traditional PFP→performance research and in estimation of the utility of PFP is the nature of the performance distribution. Performance has been assumed, explicitly or implicitly, to be normally distributed, including in utility analysis. However, in at least some jobs, performance appears to be nonnormally distributed such that there are some individuals with extraordinarily high performance in the upper tail who may create extraordinary value for the organization that is not well-captured in current models (and that is not necessarily compensated accordingly). For example, if performance is measured using only a 1 to 5 rating scale where 5 is the highest possible performance, and in a case where the rating scale is applied in a reliable and valid manner, the five performance rating category could include employees who create 30% more value than the average employee in the three category as well as employees who create 400% more value than the average employee in the three category. To the degree that performance is not normally distributed, the value of selecting and retaining these extremely high performing individuals is likely underestimated conceptually, as well as in attempts (e.g., utility analysis) to quantify it (O'Boyle & Aguinis, 2012; Aguinis & O'Boyle, 2014; for a counterpoint, see Beck et al., 2014). A related literature concerns "stars" (e.g., Call et al., 2015; Groysberg et al., 2008; Kehoe & Bentley, 2021), where again certain individuals are known to have disproportionate influence on organizational outcomes.

Of particular relevance for future research is how well the pay distribution maps onto the performance distribution in the presence of high performers/stars, and what the consequences are when it does not. This mismatch in distributions could occur, as noted above, by the failure of most performance rating systems to differentiate among high performers and extreme high performers, all of whom receive the same rating. Pay system constraints can further exacerbate the situation. Merit budgets are typically set at a fixed total dollar amount, and for a given performance rating (e.g., a 5) a manager may be given a range for raise allocations to allow some discretion to differentiate among people receiving the same rating; even so, to keep total raises awarded from exceeding the budget there is always a maximum raise that can be awarded to any individual. This constrained raise amount may effectively decouple pay from actual performance for very high performers. Even in PFP systems with a more direct relationship between performance and pay, such as a piecework or commission system, again, for budget reasons, there are often caps or maximum payments that restrict the payout to any individual.

A concern that arises when pay distributions do not align with performance distributions, either due to faulty assumptions about performance distributions or due to budget constraints that limit the PFP distribution, is that valuable employees will either search for alternative organizations where pay better corresponds to their performance value and/or that these other organizations will identify such employees as being significantly underpaid and seek to poach them (e.g., Lee et al., 2008).We could envision future research linking these distributional issues to PFP, by for example, studying how PFP can be better designed to cost-effectively incentivize and retain these extraordinarily high performers while also appropriately motivating and retaining the well-performing non-stars (e.g., solid performers but not likely to become stars) who work alongside stars, and who make up the majority of the workforce.

Finally, our summary above focuses on mean effects of PFP. Of course, PFP effects can differ, depending on horizontal fit with other employment practices and vertical fit with corporate/business strategy (Gerhart & Rynes, 2003; Gomez-Mejia et al., 2014). Issues of fit can also include (Kim et al., 2022) whether PFP effectiveness varies depending on the type of work (e.g., interesting/creative or not) and type of performance (e.g., quality vs. quantity).

3.3 | PFP intensity, risks, results, and behaviors

Recall that PFP intensity refers to how large the payoff is to performance under PFP. According to agency theory, increases in PFP intensity are expected to generate gains in motivation, but the challenge is to prevent risk-bearing concerns among agents (employees) from undermining these gains. This helps explain why PFP intensity is typically stronger in higher level/paid jobs where differences in performance are more consequential (similar to SDy in utility theory, e.g., Sturman et al., 2003) and are tied to individually separable, results-based performance, but where risk bearing is more feasible due to the overall higher levels of salaries being paid. Another trade-off is that higher PFP intensity, although increasing motivation, at the same time can increase the possibility of undesired/unintended effects. The equal compensation principle (Milgrom & Roberts, 1992, p. 228) essentially says that when different performance goals are not incentivized equally, an employee will focus on those activities most incentivized in the PFP plan and ignore others. This explains many of the potential pitfalls of PFP which include (Gerhart & Fang, 2015, p. 490): "excessive risk taking, excessive competition within the firm, focusing too little on performance measures (e.g., quality, customer service, long-term performance) not explicitly included in the PFP plan, and focusing too much on (including gaming or manipulating) performance measures (e.g., sales, stock returns) that are included in the plan." All these concerns are further complicated, with behaviors and outcomes more difficult to predict, under complex PFP systems when multiple PFP types are utilized at the same time (Park & Sturman, 2016; 2022).

Most unintended, negative outcomes of PFP seem to occur with the use of individual incentives (Gerhart, 2017) or something very similar (e.g., sales commissions) where PFP intensity is strong, pay depends solely (or nearly so) on results alone (e.g., mortgage revenue generated) and the equal compensation principle is ignored by omitting performance criteria risk factors (e.g., such as risk of loan default in incentives for loan originators in the subprime mortgage crisis) or ignoring whether acceptable behaviors are used to achieve results. Including subjective (behavior-based) measures of performance and making sure pay depends on them, and not just on results, is one step to reduce the likelihood of unintended negative consequences. Likewise, reducing the strength of PFP intensity that is based on results is another.

Although it is well-known that subjectivity in performance evaluation (e.g., via typical behavior-based performance ratings) can cause its own array of problems (too much/too little performance differentiation, rater biases, lack of credibility; Pulakos et al., 2019), behavior-based performance measures continue to be a central element in PFP in organizations for the very reason that it does allow subjectivity, or to use a more positive term, judgment. In addition, results-based performance measures, especially those allowing separation of individual contributions, are simply not available for all jobs. Behavior-based measures typically are. As research in *PP* has documented, subjective and objective measures of performance overlap in part, but subjective assessments provide independent information. Heneman (1986, p. 820) advised "not treating ratings and results as substitutes" based on his meta-analysis finding of a mean corrected correlation of .27 between the two. Another study (Bommer et al., 1995) found a somewhat higher correlation (corrected mean $r = .39$), but again the overlap was limited (variance shared = $.39^2$ = 15%, thus leaving 85% unshared). Importantly, they also noted each type of performance measure has problems (e.g., objective measures can be "excessively narrow" and subjective measures can be "excessively prone to contamination" such as supervisory biases). Thus, they suggested "an integrative approach" (using both), which may not only balance out each's problems, but is also necessary because "no measure could 'objectively' measure all relevant performance aspects. Some key measures are necessarily subjective" (p. 599).

This advice has stood up well to the test of time. Indeed, recent research demonstrates the value of subjective performance evaluation providing important unique performance information that can temper downside risk of PFP based only on objective performance (e.g., by facilitating the adjustment of standards when they become unattainable; Maltarich et al., 2017) and can motivate employees to engage in behaviors by including these in performance evaluation (e.g., helping behavior, He et al., 2021; mentoring and building long-term customer relationships, Takahashi et al., 2021) that might otherwise be ignored (e.g., as in Wright et al., 1993). These work by offsetting overreliance on objective (results-based) performance measures, consistent with the equal compensation principle (Milgrom & Roberts, 1992). Of course, the benefits of using subjective (behavior-based) evaluations in conjunction with objective measures are enhanced further by taking steps to improve the construct validity and reliability of subjective measures (i.e., by having clear, agreed-upon performance dimensions and standards; use of multiple raters, Viswesvaran et al., 1996).

3.4 | Pay transparency

Pay transparency involves organizational communication along multiple dimensions, including: (1) providing information about pay outcomes (e.g., disclosing the range of pay for an employee's own job level and/or similar information on job levels above/below, or publishing a list of all employees' pay, such as is done for many state and local government workers); (2) providing information about *how* pay is determined (e.g., bonus formula or salary determination process); (3) formal or informal policies about whether employees are allowed to discuss pay among themselves, and (4) the modes and channels to use to communicate about pay (for reviews that consider these dimensions, see Arnold & Fulmer, 2019; Colella et al., 2007; Fulmer & Arnold, 2020; Fulmer & Chen, 2014; Marasi & Bennett, 2016). A lack of pay transparency is referred to as pay secrecy. Historically, in the U.S. private sector, for nonunion employees, there has been little perceived pay transparency in terms of ability to discuss pay with coworkers or ability to access company-provided information on pay, with pay secrecy perceptions continuing to be reported in recent years (e.g., 2017–18; Sun et al., 2021). In the U.S., Executive Order 13665, which amended existing Executive Order 11246 and became effective in 2016, as well as recent laws in a growing number of states and other countries, seek to change that, driven by the belief that more transparency will reduce pay discrimination by shining a light on pay inequities. Whether pay transparency is beneficial depends on the perspective (employee, organization, societal) and circumstances (Brown et al., 2022; Colella et al., 2007; Fulmer & Arnold, 2020). Consistent with the logic above, an organization with known pay inequities may decide to avoid pay transparency. An organization may also avoid transparency to limit worker mobility by limiting their information on pay, making comparisons difficult. Of course, mobility is a key to workers moving to their most productive job/organization match (Gerhart & Feng, 2021; Weller et al., 2019), so while the private return to an organization of pay secrecy may be positive, the effect on individual workers and on society may be negative (Colella et al., 2007).

Initially, employee reactions (perceptions of fairness, satisfaction, motivation, performance) to pay transparency were expected to be uniformly positive. For example, in 1967, Lawler wrote: "The total picture is one arguing strongly for organizations adopting more open policies with respect to pay" (p. 188). He reasoned that otherwise, "it seems unlikely that employees will ever see a clear relationship between pay and performance…" (p. 188) and without this, organizations would miss out on a likely "net gain in the form of increased motivation…and increased satisfaction with pay."[ii] In their review, Colella et al. (2007) more fully developed the fairness-based logic for these predictions and, like Lawler (1967), they argued that perceived fairness would be higher with greater transparency. The case for more open pay is based on evidence (e.g., Lawler, 1967; Milkovich & Anderson, 1972) that employees are inaccurate in estimating the pay of others. Specifically, peer pay was typically overestimated. In equity theory terms, that means the outcomes of a likely social comparison would be overestimated, making underreward inequity more likely. Pay in lower-level jobs was also typically overestimated, while pay in higher level jobs was typically underestimated (see Cullen & Perez-Truglia, 2022; for similar findings on underestimating pay at higher job levels). Thus, the payoff to promotion (and thus to performance, which plays a key role in promotion) would be underestimated, reducing motivation.

However, Milkovich and Anderson (1972) found that the case for more transparency was a bit more complex. First, even where a transparency practice was in place, employees continued to be inaccurate in estimating the pay of others, in the same ways as observed by Lawler (1967). Second, those most accurate in estimating peer salaries were the most (not the least) dissatisfied with their pay (caveat: $N = 14$ only in that group). They concluded (p. 300) "results refute the contention that . . .satisfaction with . . . pay and pay differentials are a direct function of perceptual accuracy of the compensation practices of an organization." This would prove prophetic. Although it does appear that employee perceptions of process fairness is higher (due to sharing of information), employee distributive fairness reactions to this information can be positive or negative (Smit & Montag-Smit, 2019). In the case of peers at the same job level, those having higher relative pay generally react more positively to transparency, while those with lower relative pay react more negatively (SimanTov-Nachlieli & Bamberger, 2021). For example, in a university system where more detailed pay transparency (pay of individuals made public) was implemented, results included employees (having pay below the median) making more social comparisons, being less satisfied, and having increased job search intentions (Card et al., 2012). Alterman et al. (2021), found in their Study 1 (a survey of working executive MBA students) that greater perceived transparency increased (decreased) turnover intentions (indirectly via effects on organizational trust) where distributive justice was low (high). In their Study 2, using archival field data, they found no effect of transparency on voluntary turnover where distributive justice was low, but found a positive effect (reduction in voluntary turnover) when it was high. To the degree distributive justice is lower for those with pay below the median, the Alterman et al. (2021) results together with those of Card et al. (2012) suggest pay transparency effects will be more positive for those with higher relative pay, and lower for those with lower relative pay.

Brown et al. (2022) reconceptualize pay transparency/secrecy through the lens of information asymmetry among senders (organizations or colleagues) and receivers (employees), and then summarize research on the impact of pay transparency on a variety of outcomes and in different contexts. Earlier we focused on perceived fairness, satisfaction, and turnover (or intentions). Turning to work motivation and performance, Brown et al. (2022) find that most studies show a positive relationship with transparency, consistent with Lawler (1967). Belogolovsky and Bamberger (2014), for example, found, using a lab experiment, a positive effect of transparency on performance, via an incentive effect, but also via a sorting effect (consistent with Alterman et al., 2021; Card et al., 2012; SimanTov-Nachlieli & Bamberger, 2021). Note that, in an organization, obtaining the benefits of this sorting effect would require coping with pay dissatisfaction (perhaps concentrated among low performers) and sufficient turnover for these employees to be replaced by higher performers (Gerhart & Feng, 2021).

There is also evidence of individual differences in how employees react to pay transparency (e.g., due to equity sensitivity, Bamberger & Belogolovsky, 2010; for a review, see Fulmer & Chen, 2014) and in the preferences of low- and higher-paid individuals for sharing their own pay with others when transparency is taken to that point (Smit & Montag-Smit, 2019). A concern for social relationships is likely a factor here (Brown et al., 2022). Compensation activation theory (Fulmer & Shaw, 2018) argues that greater pay transparency will likely enhance the effects of some types of pay (e.g., PFP) on individuals motivated by status-seeking or social dominance concerns, since greater availability of pay information helps facilitate the comparisons needed to ascertain one's relative standing in the group.

Pay transparency has links to the literature on pay dispersion (Brown et al., 2022) given that the effects of pay dispersion likely depend on whether it is viewed as fair or not, and that evaluation (and to some degree its accuracy) depends on having information about the pay of peers and other social comparisons (Shaw, 2014; Trevor et al., 2012). Likewise, research on i-deals in PFP (Abdulsalam et al., 2021; Maltarich et al., 2017) is also connected to the degree that there are offsetting effects (positive for those receiving i-deals, potentially negative for those not) driven by social comparisons, which again depend on having sufficient information on pay and performance.

3.5 | Challenges in defining and measuring PFP

A construct definition issue is that, at the individual level, PFP in organizations is often thought of as the annual merit pay exercise. Understandably, that sometimes leads to skepticism about whether there is enough PFP to be meaningful/motivating. For example, in recent years, the average annual merit increase has been 3%–4%. Survey data show that 85% of employees receive either a three or a four merit rating on a five-point scale with a three receiving an average merit increase of 2.6% versus 3.6% for a 4 (Gerhart, 2023, Chapter 10). On an annual salary of $80,000, that would amount to a difference of $800/year, $15.38/week, less still after taxes.

However, PFP is likely much stronger if observed over a longer time period, thus capturing the fact that higher performers are promoted (at their current or a new employer) to higher job levels, which, in turn have not only higher base pay levels, but also higher levels of short- and long-term incentives (Gerhart & Fang, 2014; Gerhart & Milkovich, 1992; Trevor et al., 1997). A promotion to a higher job level brings a much larger pay increase. The modal midpoint progression (% difference between adjacent job level pay midpoints) is 15% (WorldatWork & Deloitte, 2019). Thus, for example, someone receiving 3 promotions within a few years would increase their pay (beyond merit increases) by $1.15 \times 1.15 \times 1.15 = 1.52\times$ (Gerhart, 2017). Further, we know that performance is linked to promotion. For example, Gerhart and Milkovich (1989) reported that a 1-point higher performance rating on a 4-point scale was associated with 48% more promotions over a 6-year period, relative to those with a mean performance rating. In addition, short-term incentives such as merit bonuses, which have grown substantially in use (Gerhart & Fang, 2014), are often more strongly linked to performance than merit raises and likewise can have stronger effects on performance (Nyberg et al., 2016). In summary, PFP can be conceptualized and measured over different time periods and with different measures of pay. PFP will be stronger to the degree the time period is longer than a single year and to the degree that the measure of pay includes not only annual merit pay increases, but also short-term and long-term incentives (and promotion-based increases). Further, measuring PFP over time as how the magnitude of percentage pay growth differs as a function of performance over that same time period offers PFP information beyond knowing whether simply knowing whether an employee is covered or not by different types of PFP plans.

3.6 | Pay dispersion

Another construct definition issue concerns pay dispersion, particularly among individuals doing similar jobs (i.e., horizontal dispersion), and what it adds beyond PFP. Pay dispersion refers to variation/differentiation in pay or unequal pay among members of a group. The most basic theoretical use has been to suggest that more pay dispersion makes sense when required interdependence/cooperation is limited (vs. substantial, as in teams). The argument is that less pay dispersion/inequality is more equitable and thus more likely to foster cooperation. Shaw (2014, p. 524) describes this framing as “a dramatization” that got people’s attention, but “a significant oversimplification.” Simply put, equality ≠ equity. Indeed, under equity theory, we would expect an employee with higher contributions (e.g., performance) to perceive inequity (i.e., feel under-rewarded) if his/her pay is the same as (rather than higher than) an employee/social comparator having lower performance. This is consistent with research reviewed above that performance is generally viewed as the most legitimate basis for pay.

In line with this logic, research on pay dispersion has determined that *how* the dispersion in wage rates arises matters. To the degree that dispersion is caused by use of PFP, which is commonly the case, it is seen as more legitimate (versus when due to non-performance factors, such as organizational politics), with more positive/less negative consequences (Kepes et al., 2009). Dispersion, when created via use of PFP, has been associated with better performance, less turnover and enhanced attraction of high performers (Shaw, 2014; Trevor et al., 2012). Several excellent comprehensive reviews of pay dispersion already exist that elaborate on this and on other contingencies; in the interest of space, we refer the reader to those for further information (Conroy et al, 2014; Downes & Choi, 2014; Gupta et al., 2012; Shaw, 2014).

As noted earlier in discussing the potentially mixed effects of enhanced pay transparency, we and others (e.g., Fulmer & Li, 2022) anticipate a complex interrelationship among pay dispersion, PFP, and pay transparency. Greater transparency enhances the link between expected performance and pay, which would normally be a positive for performance outcomes, but transparency also enhances the opportunity for social comparisons. Other research on ease of social comparisons (that do not involve transparency per se, but proximity) finds that under some circumstances, easier social comparisons can result in negative effects, offsetting some of the positive incentive effects of performance-based pay differences (e.g., Obloj & Zenger, 2017). The research on this is relatively recent and evolving, so it remains to be seen whether and under what conditions the positive effects of PFP-caused dispersion are affected by greater pay transparency.

3.7 | Endogeneity in compensation research

Earlier, we summarized how researchers address (or do not address) endogeneity in compensation research. Endogeneity exists when the covariance is nonzero between the exogenous variable (compensation here) and the regression equation error term (e.g., Wooldridge, 2002, p. 50). When endogeneity is present, using ordinary least squares regression results in a biased estimate of the compensation regression coefficient. Sources of endogeneity are omitted variables, simultaneity, selection bias, and measurement error (Hill et al., 2021; Wooldridge, 2002).

Simultaneity could be a concern in compensation research, for example, if in response to high employee turnover, wages are raised and/or a retention incentive is introduced. Without addressing the endogeneity, cross-sectional data might indicate a negative relationship between the compensation programs intended to reduce turnover, without revealing that, in fact, turnover would have been (even) higher without these programs. Another simultaneity challenge is determining whether pay → performance or performance → pay (or whether it is both). An example is the Trevor et al. (2012) study, which used instrumental variables to test the robustness of their finding that team compensation (pay dispersion based on performance) → team performance. As another example, in implementing a PFP program to improve performance, if other factors in the organization change at the same time (e.g., product demand, technology, workforce quality), the effect of PFP may be over/underestimated. Or, again, PFP may be a response to remedy subpar performance.

The gold standard for dealing with endogeneity through design is the experiment (Podsakoff & Podsakoff, 2019), which has several defining features, including (a) "variables are manipulated and their effects upon other variables observed" (Campbell & Stanley, 1963, p. 1); (b) random assignment, when executed successfully (e.g., sufficient sample size, limited differential attrition), creates equivalent experimental and control groups; (c) time precedence; and (d) it allows "measuring actual behavior" (e.g., in contrast to a survey asking about a hypothetical situation) (Colquitt, 2008, p. 618; see also Dipboye & Flanagan, 1979, Table 5). As we saw earlier in Table 1, experiments have been used regularly in research on the effect of PFP on performance over time. More common in field settings, the quasi-experiment (Campbell & Stanley, 1963) refers to manipulation of (exogenous) variables, but without random assignment (often not feasible in the field). Strength of causal inference depends on the design. As we also saw earlier in Table 1, quasi-experiments have not been used in research on PFP → performance since 2010.

The second general approach to endogeneity uses statistical modeling during data analysis (Hill et al., 2011), most commonly using instrumental variables (e.g., Maydeu-Olivares et al., 2020), propensity scores (Harder et al., 2010; Rosenbaum & Rubin, 1983), fixed effects (to control time-invariant omitted variables) or modeling the selection process (Heckman, 1979).Use of control variables is another avenue, but does not generally address sources of endogeneity such as simultaneity, selection bias, or measurement error.

In Tables 1 and 2, we saw that two studies used fixed effects, one used a Heckman (1979) correction for selection bias, and one used instrumental variables. The latter (Trevor et al., 2012) was in the study of how team (horizontal) pay dispersion influenced team performance. Another example from outside the scope of our review (from the accounting literature, Rouen, 2020) likewise used instrumental variables to study the effect of pay dispersion on performance (here, vertical dispersion and at the organization level). Notably, it also used propensity score matching, which has not been used to date in the compensation literature we reviewed. Although we have focused primarily here on statistical approaches to endogeneity because they have been especially rare, a focus on design approaches in field research is also encouraged, including, but not limited to designs such as regression discontinuity and difference in differences (Hill et al., 2021; Lechner, 2011; Lee & Lemieux, 2010). In summary, we encourage researchers to do more to address endogeneity challenges.

3.8 | Country differences in compensation research

Earlier we saw that only three studies in Table 1 used data from multiple countries and only one of the three used between-country variance to examine a substantive question. Thus, country differences, including the key question of whether compensation programs work better/worse in some countries (as indicated by a statistical interaction), have been neglected. Multilevel modelling (e.g., Raudenbush & Bryk, 2002) is useful to address these questions because it effectively handles nesting of (dependent) observations within countries and also focuses attention on the need to decompose/explain country effects using specific country characteristics (See example in Rabl et al., 2014, using a system of employment practices, including compensation). This decomposition is crucial becausecountry effects have often been misinterpreted as national culture effects (e.g., Bhagat & McQuaid, 1982; Gerhart & Fang, 2005; Rabl et al., 2014). Rather, the country effect is an upper bound on the effect of any specific country characteristic, of which there are many (e.g., labor market flexibility), in addition to national culture.

The primary focus of international work on pay has been on whether reward allocation criteria differ by country (Gerhart, 2008). A key theoretical logic (Early & Erez, 1997, p. 74) is that “the equity principle [reward based on performance] is commonly used in individualistic cultures [like] the United States” in contrast to the “quite different” principles (e.g., equality) used in cultures such as in China. Zhou and Martocchio (2001, p. 115), for example, report “compared with their American counterparts, Chinese managers… put *less* emphasis on work performance when making bonus decisions [and] put *more* emphasis on personal needs…” [emphases in original]. This conclusion was based on statistical significance. Importantly, the authors also reported that individual performance explained 64.2% of the variance in bonus allocation decisions, whereas the interactions (i.e., the degree to which respondents from China and U.S. used different allocation rules) explained only 1.1% of the variance. (Note also that national culture was not actually measured.) Thus, the key takeaway is arguably that performance dominates reward allocation decisions in both countries. More broadly, a meta-analysis by Fischer and Smith (2003) examined the degree to which the use of the equity (performance) rule versus the equality rule varied by the national culture dimension of collectivism, using data from 14 countries. They reported a mean $r = .065$ (i.e., an $r^2 = .0042$). Thus, national culture had no meaningful effect.

3.9 | Diversity and inclusion in compensation research: Pay equity

Discrimination is typically divided into (a) access discrimination, where members of a group are disproportionately denied opportunities to work in certain occupations, industries, jobs, and/or firms and (b) valuation discrimination, where members of a group are paid less despite working in similar positions and having similar attributes. Pay inequity most directly reflects valuation discrimination. U.S. Census Bureau data on median earnings of full-time, year-round workers shows that women earn 82% of what men earn (up from 60% in 1980). Compared to white men, Black, Hispanic, and Asian men earn 81%, 74%, and 131%, respectively. Compared to white women, Black, Hispanic, and Asian women earn 90%, 77%, and 124%, respectively. Blau and Kahn (2017) estimate that 62% of the female-male wage gap is explained by differences in work experience (years, continuity), occupation, and industry. The unexplained part (38%) is typically taken to reflect discrimination. If women have experienced access discrimination, thus being underrepresented in some occupations and industries, 38% is an underestimate. There is also an economics literature on race-based discrimination (For reviews, see Lang & Lehman, 2012; Lang & Kahn-Lang Spitzer, 2020; Neumark, 2018).

Within organizations, the term "Glass Ceiling" identifies women's lack of representation at higher job (pay) levels as a key factor in their lower pay. For example, Catalyst (2020; see Gerhart, 2023, Exhibit 17.15), reports that although women are 45% of employees in S&P 500 companies, they are 37% of first/mid-level officials/managers, 27% of executive/senior level officials/managers, 21% of Board seats, and 6% of CEOs. This reinforces the importance of examining the role of promotion in compensation research.

4 | PRACTICAL IMPLICATIONS AND FUTURE RESEARCH RECOMMENDATIONS

In this paper, we undertook two tasks. First, using over 30 years of articles in *PP*, *JAP*, and *AMJ*, we examined the current state of as well as trends in how research on compensation → performance is theorized, its most studied aspects, and designs/methodologies employed. Given that we found that the most-studied compensation construct has been PFP, and given the long-held view of PFP's strategic importance to organizations, we used the rest of our paper to focus on PFP, but casting a wider net than the three journals we used to examine the state of the field and trends. Our aim was to highlight key decision areas in the design of PFP and bring to bear theory (broadly defined) and evidence to better understand how PFP works, including trade-offs and contingency factors. In doing so, we included work on performance measurement/PFP design, pay transparency, and pay dispersion.

Based on this review, we identified several important takeaways from existing research that are of current and practical relevance for organizations and compensation managers. We also offer a set of recommendations for future scholarly research on employee compensation.

4.1 | Practical implications for organizations

A summary of practical implications and recommendations emerging from this review is found in Table 3. Given the focus of this review on PFP, our first category of recommendation focuses on the pay aspect of PFP. We advise organizations to be aware of the effects of PFP intensity, not only because of its importance for achieving desired performance outcomes, but also to anticipate the potential need for mechanisms to address the risk of negative consequences for high-intensity PFP programs. All else equal, PFP programs that offer stronger payouts for performance (i.e., with a strong pay-for-performance link and that are large as a percentage of overall pay) need to be monitored more closely than those with smaller payouts. Another concern that has received much attention, but which we did not focus on in this review, is that higher PFP intensity potentially harms intrinsic motivation and performance in work that is interesting and/or requires creativity (Deci et al., 2017; Weibel et al., 2010). This concern is more important to the degree such work is becoming increasingly important to economic well-being. However, our reading of the literature is that the evidence (and most theory) does not support this concern. (See Gerhart & Fang, 2015; Jenkins et al., 1998; Kim, 2022; Kim et al., 2022). Nevertheless, there is always a role for research that examines how best to design PFP to support all aspects of motivation, including intrinsic/self-determined motivation, and all types of performance (Gagné & Deci., 2005; Kim et al., 2022).

The performance aspect of PFP continues to be an issue requiring close attention as well. We recommend that organizations view PFP from both a short- and long-term perspective and communicate it as such to employees. It

TABLE 3 Summary of practical implications for organizations

Pay Attention to the *Pay* Component of Pay-for-Performance to Better Support Organizational Goals and Strategy	
PFP intensity—Expect (and build in mechanisms to address) greater risk of unintended consequences as PFP intensity increases	The choice of how strong the payoff is to performance involves trade-offs, pitting increased motivation created by greater PFP intensity against increased risk of unintended consequences. All else equal, PFP programs that offer more intense payouts or with graduated payouts that increase with performance should be monitored more closely than those with smaller payouts.
Pay Attention to the *Performance* Component of Pay-for-Performance to Better Support Organizational Goals and Strategy	
Design PFP and communicate performance expectations so that it is clear how employees' performance leads to short- *and* long-term pay increases.	If the goal is to better motivate employees by strengthening PFP strength/intensity, it is necessary to think beyond the annual within-pay grade merit increase program based on supervisor-rated overall performance. Other, more strategically targeted aspects of compensation (e.g., bonuses tied to specific performance goals) may be more strongly linked to desired performance. Also, over time, the major pathway for performance to lead to higher pay is promotion/advancement to high job (pay) levels. For these facts to influence employees, they must be communicated effectively.
Performance measurement – Consider more holistic (i.e., multiple) measures	A key means of reducing risk of unintended consequences is to introduce the use of behavior-based performance measures (e.g., a judgment/performance rating) to take into account unexpected factors and to help ensure employees achieve results using appropriate means/behaviors.
Is Your PFP System Attracting and Retaining the Desired Talent?	
Factor in sorting effects more explicitly when designing PFP programs	A major change in conceptualizing how PFP influences performance is the relatively recent recognition of how PFP affects not only what current employees do (incentive effect), but also which employees choose to join/remain with the organization (sorting effect). High performers, on average, prefer stronger PFP strength/intensity. Ignoring sorting effects results in a significant underestimation of the PFP consequences.
Check your PFP system constraints	For organizations that are highly dependent on "stars" or extremely high-performing individual contributors, it is important that the payout distribution of the PFP system is adequate to be able to reward the extreme performance of top performers.
Be careful about making exceptions	For a particular employee it may *seem* helpful to make exceptions for below-standard performance, but keep in mind that exceptions weaken the perceived performance-pay linkage of other peers within the unit and threaten the effectiveness of the whole PFP system as a motivational tool.
What Do We Know So Far About Pay Transparency Effects on Performance?	
Pay transparency—It's complicated. . .	More pay transparency does not necessarily lead to better performance and more positive employee reactions. Research has identified a number of contingency factors, including current pay level.

is important to think beyond annual merit pay, to incorporate individual incentives as appropriate (e.g., bonuses tied to specific strategic performance goals), and to emphasize pathways to promotion-based pay increases. We also recommend more holistic approaches to performance measurement that incorporate behavior-based performance (in addition to results-based performance) as a way to take into account unexpected factors and ensure that appropriate means are used to achieve objective performance metrics. Consider Novartis as an example. It reached a

$678 million settlement with the U.S. Justice Department over improper incentives to persuade doctors to prescribe Novartis drugs (Vigdor, 2020). In response to this, as well as other unethical sales/bribery practices overseas, Novartis changed their incentive system such that payouts would no longer depend solely on results, but also on a rating of whether employees exhibited appropriate values and behaviors in achieving these results. Novartis also decided to reduce PFP intensity by capping the bonus to a maximum of 35% of compensation (Liu, 2018).

Our third set of recommendations center around the issue of how PFP systems can best attract and retain the desired talent. PFP systems are often designed with the motivation of current employees in mind, but we recommend that organizations step back and take a broader view that considers sorting. Otherwise, they are likely to underestimate the consequences of PFP adoption, given that high performers, on average, prefer organizations that offer stronger PFP programs. Another recommendation is to check that the constraints (e.g., budget controls, payout caps) built into the PFP program are not artificially or unnecessarily undermining the goals of the program. This is especially important in organizations that are highly dependent on "star" employees or extremely high-performing individual contributors, where it is important that the payout distribution of the PFP system is adequate to reflect the performance distribution, such that high performers are appropriately compensated. And finally, we note that research (e.g., Abdulsalam et al., 2021) has found that making exceptions ("i-deals") for below-standard performance in a PFP system can create significant negative reactions among peers. Although it may be helpful to individual employees who are struggling, if they are outnumbered by peers, then any such negative peer effects will dominate, translating into lower unit performance.

And finally, we note that the research on pay transparency is relatively new but beginning to accumulate quickly. As of now, the message is mixed, which suggests that organizations contemplating voluntarily increasing transparency should proceed carefully. Our review highlighted that more pay transparency does not necessarily lead to better performance or to more positive employee reactions across the board. Research has identified a number of contingency factors, including current pay level, and researchers are still in the midst of understanding how those factors affect the consequences of increasing transparency.

4.2 | Recommendations for future research

We summarize four categories of recommendations for future research in Table 4. We begin with attention to the theoretical foundations of the field as they have evolved in the studies in this review. As noted earlier, a few well-established theories (i.e., equity, expectancy and tournament/sorting theories) dominated throughout the time frame of our study, including during the 2010–2022 period, while others commonly cited in earlier decades have fallen by the wayside recently. Notably, we observed the decline in the use of agency theory and goal-setting theory, with none of the studies of employee compensation in the 2010–2022 period using them (Table 1). Both were frequently used pre-2010 and supported empirically, and have continued to develop theoretically in the literature (e.g., Locke & Latham, 2002; 2020). Although new phenomena sometimes require new theories, we have seen examples of how classic compensation/motivation theories can be extended quite fruitfully to understand contemporary research questions, and we encourage future researchers to give some thought to such application. For example, the emerging line of research on the effects of pay transparency on performance and sorting could be informed by attention to how social pressure and comparisons facilitated by pay transparency influence individuals' acceptance of assigned goals or the setting of self-set goals and subsequent goal attainment; such research would be particularly valuable in settings where social comparisons are likely and/or encouraged, such as in co-located sales groups or workgroups that are highly connected via digital work platforms or social media. Intriguing research in the area of goal setting has also found (in academic settings) that writing about goals enhances both performance and persistence (Morisano et al., 2010; Schippers et al., 2020; for a brief review see Locke & Latham, 2020). The mechanisms for this are still to be determined, but given that people now routinely document personal and professional goals digitally via writing on social media, as well as in videos, podcasts, etc., it would be interesting to examine whether digital documentation of employees' goals in these

TABLE 4 Recommendations for future research

Theoretical Foundations	
Consider well-established theories for studying new phenomena	Goal-setting, agency, expectancy and equity theories have a strong foundation of empirical support. While some may assume that new phenomena must call for new theories, a hallmark of good theory is that it is useful for understanding what seem to be novel phenomena. And indeed, established theories can often be extended and refreshed through applying them to new phenomena.
Understudied Aspects of PFP	
PFP strength/PFP intensity	PFP is not simply dichotomous. Its strength/intensity is important to include to fully understand its effects.
PFP payout holdbacks or clawbacks	Some employee PFP systems (e.g., group gainsharing plans), pay for recruitment referral programs, key employee retention programs (e.g., during a restructuring) and executive PFP programs include provisions to delay or recover some or all of a PFP payout until a later time to account for contingencies and try to align employee/executive decisions with long-term best interests of the organization. Agency theory-based research on the incentive and sorting effects of holdbacks and clawbacks would be valuable.
Performance measurement	Results- and behavior-based PFP plans can have different effects and most organizations use a combination. However, research using both is rare. Research including both is needed.
Performance effects of the most common form of PFP, merit pay?	Meta-analytic work on PFP effectiveness, as well as ongoing laboratory work, focuses on individual incentive plans (e.g., bonuses). These types of plans, at least in pure form, are rare. Merit pay continues to be very widely used (and merit bonuses have grown in use). There is proportionally much less empirical research on these plans. More such work would be valuable to understand their incentive and sorting effects.
Extreme performance distributions: Effects on pay systems	Research needed on how PFP can be designed to cost-effectively incentivize/retain extraordinarily high performers/stars while also appropriately motivating and retaining the well-performing non-stars who make up most of the workforce.
Pay dispersion	To the degree pay dispersion continues to be a starting point for some compensation research, it would be useful to clarify what it adds beyond studying PFP dispersion. A fundamental requirement of PFP is creation of pay dispersion that intentionally corresponds to performance/contribution dispersion; research to date has found that overall dispersion's effects are largely tied to how performance-based it is.
Pay Communication and Transparency	
Effects on performance and on diversity and inclusion goals	Greater attention needed to contextual factors (individual, situation) that determine whether different pay communication/transparency strategies improve motivation/performance, reported satisfaction/fairness, and sorting. It is also important to understand whether employee reactions to increasing transparency versus to decreasing transparency are symmetric (i.e., how challenging is it to reduce or roll-back pay transparency once it has been increased?)
Effects on diversity, inclusion and equity	More work is needed to understand the degree to which increased pay transparency reduces pay inequities within organizations, a commonly stated goal for transparency. If it does, through what means is this accomplished (e.g., increasing pay of underpaid groups or capping/constraining pay of higher paid groups) and at what overall cost to an organization?
Pay information availability effects	Both increased transparency on the part of organizations and the quantum leap in the easy availability of pay information on the Internet have increased how informed employees are about alternative opportunities. How does this influence employee mobility, their performance, and their other reactions? It is often assumed that greater pay information availability facilitates more pay comparisons among employees, but research so far suggests that this does not always happen. What individual and situational factors strengthen/weaken the link between information availability and social comparisons?

(Continues)

TABLE 4 (Continued)

Methodological Concerns	
Endogeneity	In field work, statistical approaches continue to be rare and design approaches have become rare.
International samples	Few studies use samples from more than one country, and none examine whether PFP effectiveness varies across countries; more research is needed to understand whether PFP works differently in different parts of the world.

ways have similar positive effects on performance and persistence in PFP settings. In other words, could sharing PFP-related performance goals in these digital and public ways enhance the effects of PFP through enhancing cognitive focus (i.e., via visualization of an "ideal future," as Morisano et al., 2010 suggested), through a sense of accountability to one's audience, or both? Might some employees proactively use social media in this way to self-regulate and support their own goal attainment? Although it may seem that new phenomena call for new theories, a hallmark of well-developed and established theory is that it often remains quite useful for understanding apparently novel phenomena. And indeed, established theories are reinvigorated and extended through such application.

The second category of recommendations highlights several under-studied aspects of PFP that we noted in our review. First, more research is clearly needed on the construct of PFP intensity and its effects. Relative to other aspects of PFP (e.g., objective vs. subjective performance rating, individual vs. group performance basis), we know less about PFP intensity effects on performance and other outcomes, and about individual and situational contingencies that moderate these relationships. We recommend that research go beyond conceptualizing and operationalizing PFP as either present or absent, and develop research that theoretically and empirically take into account the magnitude of the pay-performance relationship in terms of how sensitive payouts are to performance and how large actual or potential PFP payouts are relative to employees' overall pay. Second, one recommendation for reducing the risk potential for negative consequences of a strong incentive is to delay its full payment until there is greater certainty that the performance has actually been delivered as expected (a "holdback") or to include a provision where previously paid incentive must be returned (a "clawback"). Clawbacks are commonly used in executive pay situations, although they could be used in other cases as well, while holdbacks or similar delayed payments are used in group-based pay systems (e.g., gainsharing) or for special payments such as employee referral incentives that do not pay out fully until the referred employee reaches a particular tenure benchmark (Pieper, 2015). In our review, we noted no research that examined whether such mechanisms are used or are effective in employee PFP situations, but we believe such work would be fruitful. Third, as noted previously, results- and behavior-based PFP plans are often used together in organizations. Research on complex PFP plans that use both is rare, and more would be valuable. Fourth, the most common form of individual pay-for-performance plan in use in organizations is the merit pay plan, and pure individual incentive plans are rare. However, there is far more research on individual incentive plans. In line with previous reviews (e.g., Rynes et al., 2005), we think more work on merit pay plans would be valuable to better understand their incentive and sorting effects. Fifth, we recommend that more research be undertaken to understand how organizations structure their pay systems when they are faced with extreme performance distributions (i.e., a few star employees or individual contributors with exceptionally high performance together with a majority of well-performing non-stars whose contributions are also needed). And finally, we note the overlap in research on pay dispersion with research on performance-based pay, with growing evidence that dispersion's effects are largely driven by whether dispersion is attributable to performance-based pay differentiation. We recommend that future pay dispersion research consider what it can add that goes beyond what we know about the effects of performance-based pay.

A third category of recommendations relates to growing interest in pay communication and transparency. Research to date suggests that more transparency does not necessarily lead to better performance and employee reactions. Greater attention is needed to situational and individual factors that determine how different communication/transparency strategies influence outcomes. One important question for organizations is whether they can adjust transparency if they find it is not working as planned. Are employee reactions to reductions in transparency of similar magnitude as reactions to increases in transparency, or might we see more asymmetrically negative reactions to reductions in transparency? Given the commonly stated assumption that greater transparency will contribute to reducing pay inequities, more research is also needed to understand whether this is actually happening. If it is, through what means is this occurring (e.g., through raising pay of lower-paid groups or capping/constraining pay of higher paid groups or both) and what are the effects on different employee groups' performance and turnover and on overall (unit) performance and costs? One final area relates to the effects of increased pay information availability. More work is needed to understand how the vast array of pay information that is now publicly available on the internet is influencing social comparisons among employees as well as workers' information about alternative opportunities. What are the effects on employee mobility, performance, and other reactions?

In terms of methodological recommendations, we found substantial diversity in design/methodological approaches regarding how performance is measured, level of analysis, and whether substantive variance in compensation is generated using within- or between-organization designs. However, we saw at least two methodological concerns that give pause. One is that attention to internal validity and endogeneity challenges in the field seem to be limited, especially the use of statistical approaches; we also noted that the use of design approaches (e.g., quasi-experiments) to addressing endogeneity have waned. Second, we found no work using multi-country samples on whether compensation and/or PFP specifically is differentially effective in different country contexts. (See Gerhart, 2008, for a review of the work that comes closest.) We recommend greater research attention to both of these issues. Utilization of multi-country samples, in particular, would also enable expanded attention to a richer set of cross-cultural and international issues in pay research, including comparisons of PFP incentive and sorting effects by country, country differences in utilization and effectiveness of job- versus person-based pay, and pay transparency effects across countries. Given the number of organizations managing global workforces and multinational organizations, such research would be of particular relevance to organizations, managers, and HR/compensation practitioners.

5 | CONCLUDING THOUGHTS

On the occasion of this special anniversary issue of *Personnel Psychology*, we have welcomed the opportunity to reflect on the most recent 30 years of compensation research in *PP* and other top journals. We conclude by observing, as have others over the years (e.g., Gupta & Shaw, 2014), that employee compensation continues to be an under-studied topic despite its clear importance to employees' lives and its economic significance for organizations and society. The proportion of academic research on compensation in comparison with other HR topics has been lower than the relative proportion of interest in compensation expressed in HR practitioner journals (Markoulli et al., 2017), signaling the need for more academic research to match practitioner interest in the topic. Looking forward, we hope that by the time the 100th anniversary of *Personnel Psychology* rolls around, this review and the recommendations that have emerged from it will have served to invigorate and inspire more of the important research needed to fill these gaps and better inform how organizations design and manage employee compensation systems.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

ORCID

Ingrid Smithey Fulmer https://orcid.org/0000-0003-1543-6483
Ji Hyun Kim https://orcid.org/0000-0002-9833-3991

ENDNOTES

[i] An example of a results-based individual plan is an individual incentive or sales commission. Results-based, aggregate level plans include gainsharing, profit sharing and stock plans. Merit pay is an example of behavior-based, individual plan.

[ii] Although we are not able to cover benefits, an important study documents the degree to which employees undervalue their benefits (Wilson et al., 1985).

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